

CEL: Comprehensive Counterfactual Explanations Library and Benchmark

One protocol. Multiple CE paradigms. Measurable trade-offs.

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Code & project

Benchmark scope

18
datasets

14
methods

3
paradigms

2
backbones / task

5
folds

Source · Datasets · Methods · Full-Results

Method or setup?

Different protocols obscure fair CE comparisons.

Data splits Preprocessing Predictors Constraints

Metrics

Source · Benchmark

Three contributions

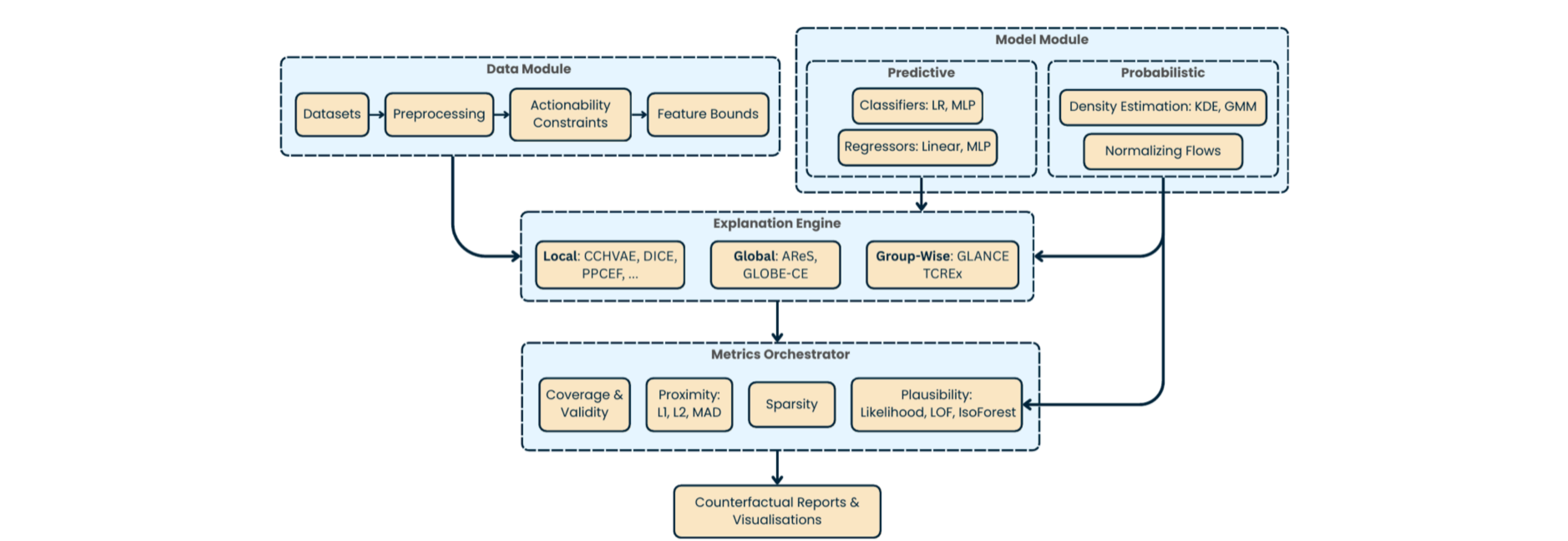
CONTROL Controlled evaluation protocol

COVER Broad benchmark across CE paradigms

EXTEND Extensible open-source benchmark workbench

Source · Introduction

One evaluation framework

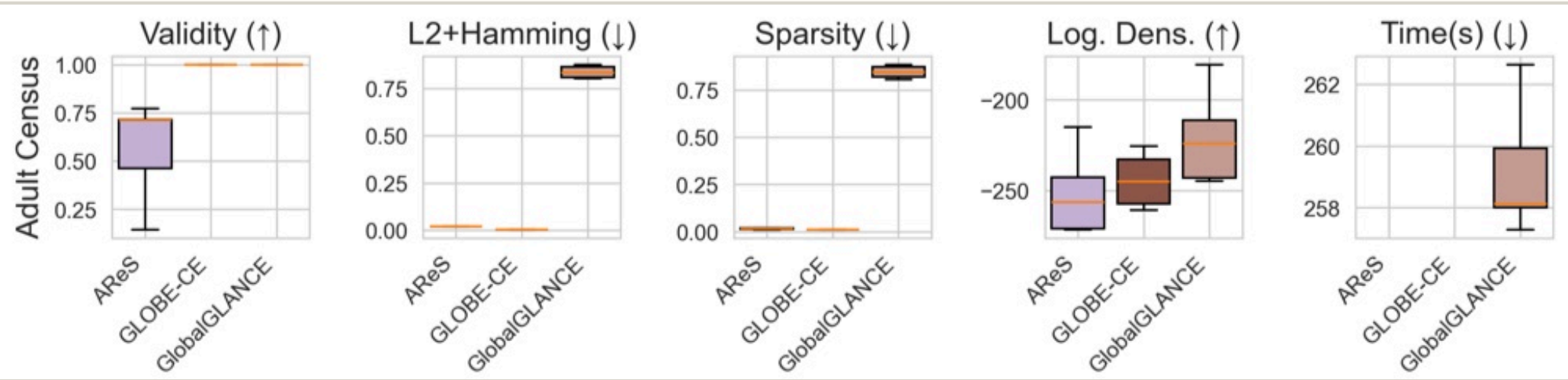


SHARED EVALUATION PROTOCOL

Source · Introduction

Global: validity and failures

GLOBE-CE and GlobalGLANCE: stronger validity. AReS: frequent missing outputs.

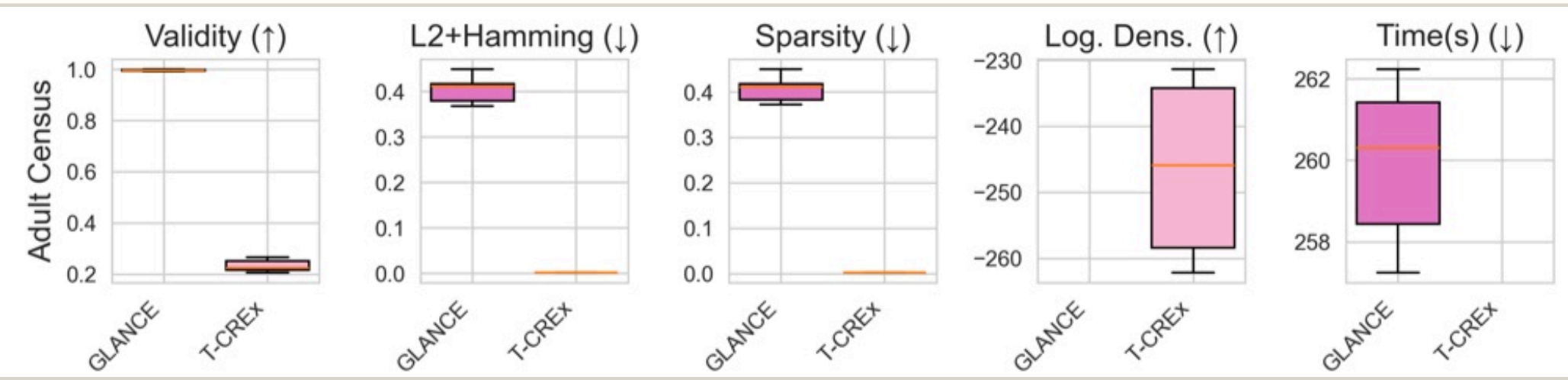


ADULT CENSUS · GLOBAL METHODS

Source · Results · tab:global_metrics_mlp

Group-wise: reach versus change

GLANCE reaches more; T-CREx makes smaller, more plausible shifts when applicable.

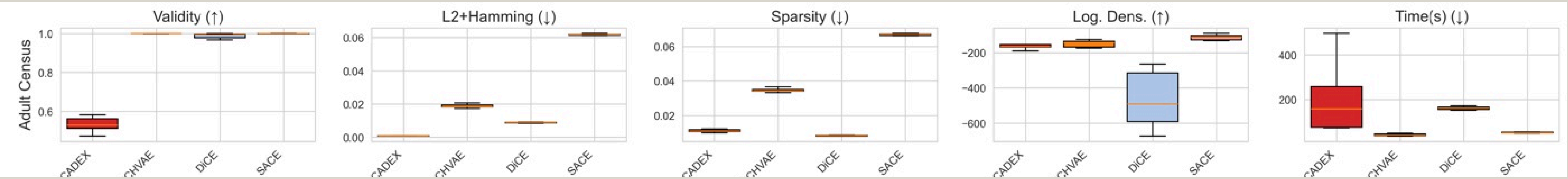


ADULT CENSUS · GROUP-WISE METHODS

Source · Results · tab:group_metrics_mlp

Local: no single winner

Validity, proximity, plausibility, and runtime trade off.



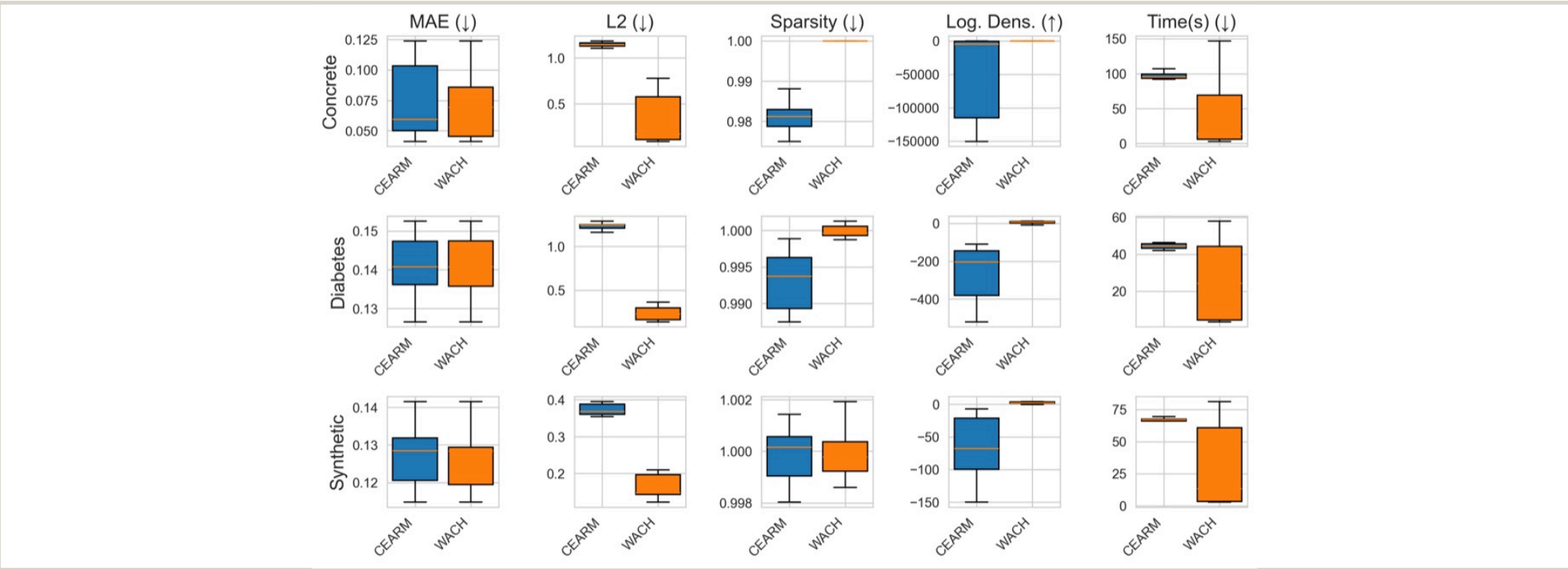
ADULT CENSUS · LOCAL METHODS

Sparsity direction unresolved; no ranking.

Source · Results · Conclusions · Metrics

Regression: beyond target error

Wachter: similar error, smaller changes, stronger plausibility, lower runtime.



CEARM AND WACHTER · REGRESSION

Source · Results · tab:regression_all_dnn